ISMIP7 Protocol Cheat Sheet – April 2026 - Following workshop

This cheat sheet will be replaced with the ISMIP7 protocol, which will include more information on the sampling of uncertainty, how CMIP models are selected etc. The aim of this short document is to provide information on how to retrieve and implement the "trial protocol dataset" and start the core simulations, so that modelers can provide feedback on the proposed implementation.

Trial protocol model

CESM2-WACCM historical and SSP585 to 2300 has been chosen as the CMIP6 model for the trial protocol. Datasets for the historical (1850-2015) and SSP585 (2015-2300) have been prepared by the ISMIP7 focus groups.

Core and sensitivity experiments overview

ISMIP7 has a set of core experiments that each group needs to perform and a series of optional sensitivity experiments. The core experiments are a compulsory set of 11 experiments with 2 ESMs, 3 projections (SSP126, SSP370, SSP585), historical and control simulations. The last experiment is the Observational Constrained Experiment (OCX). These core simulations should be made with the same set of standard parameters.

Core Exp	Scenario	Start year	End year	ESM
1	Historical	≥1850	2014	CESM-WACCM
2	Historical	≥1850	2014	MRI-ESM2-0
3	SSP370	2015	2100	CESM-WACCM
4	SSP370	2015	2100	MRI-ESM2-0
5	SSP126	2015	2300	CESM-WACCM
6	SSP126	2015	2300	MRI-ESM2-0
7	SSP585	2015	2300	CESM-WACCM
8	SSP585	2015	2300	MRI-ESM2-0
9	CTRL2015	≥1850	2300	CESM-WACCM
10	CTRL2015	≥1850	2300	MRI-ESM2-0
11	OCX	1990-2015	2025	-

The optional sensitivity experiments come under 2 distinct type:
Earth System Model (ESM) Experiments: this suite of experiments will be based around SSP370 using different ESMs. Other scenarios may be added, including CMIP7 forcing when available.
Perturbed Parameter Ensemble (PPE) Experiments: this suite of experiments will explore the impact of parameter uncertainty using one or more experiments.

- Uncertainty associated with climate forcing (ie basal melt, etc). These will typically not require a new initial state.
- Uncertainty associated with ice physics (ie: ice rheology, sliding, GIA etc). These may require a new initial state to avoid shock of instantaneous change in physics.

ISM groups can participate in ESM experiments and not in PPE experiments (and vice versa). Focus groups (FG) are proposing experiments to explore uncertainty and ISM groups are encouraged to liaise with the FG leads and the scientific steering committee (SSC) for the design of their individual ensembles.

ISM groups are to identify the relevant parameters, their range and are free to design their own ensemble. Parameters in the IPCC *very likely* range should be explored, such that the true value of parameters has 90-100% likelihood of falling in the selected range (see Mastrandea et al., 2011, Climatic Change, doi: 10.1007/s10584-011-0178-6).

These modeling choices will be recorded via a combination of readme files and experiment spreadsheet mapping. ISMIP7 will give examples. The submission requirement may be different for core and sensitivity experiments (we are considering a reduced submission of scalar time series and full 2d output at the start and end of each experiment for a large ensemble).

ISM groups are encouraged to publish the results from these experiments with their own model only (i.e. no comparison to other models) and before the IPCC deadline if desired. Analysis of multiple models will be led by FGs or in coordination with the SSC.

Initial conditions, control run and parameter settings
ISM are free to use pre-existing initial conditions for any time between 1850 to 2014, and are free in the choice of parameters. ISMIP7 provides a MIPkit with datasets that modelers may find useful, however, you do not have to use these data and are welcome to use additional datasets not included in these MIPkits. (Greenland and Antarctic MIPkit are ready). Additional datasets are suggested in the Tables presented during the November 2025 webinar.

In ISMIP6, modelers performed a single historical run and projections all started from the same state in 2015, regardless of the CMIP model used. ISMIP6 used a control run to remove any drift or historical memory associated with the historical run. With ISMIP7, the protocol is closer to CMIP7, namely that ice sheet modelling groups will perform a historical simulation per CMIP model. We will not remove control runs from projections but these are needed to assess model drift. Note that the start time of the projection for CMIP7 will be different from CMIP6, so for the final protocol we may start projections at a time that is consistent with CMIP7. The duration of the experiments depends on the CMIP forcing availability, and thus generally ends in year 2100 or 2300.

The control run is an unforced simulation, with constant climate corresponding to a 30 year climatology. This constant climate control experiment starts at year 2015 and ends in 2300. Fracture/ice shelf collapse/calving/GIA etc set constant to end of 2014 conditions (if applicable to your model). The climatology (2000-2029) should be created using a combination of the historical and the SSP126 simulations or use the datasets provided for this experiment.

The Observational Constrained Experiment (OCX) is designed to provide ISMs with best available forcing. The aim is to understand errors introduced by forcing with ESM. Instead the forcing is derived from resanalysis products for atmosphere and Greenland ocean. For Antarctic oceans scenarios are based on expert understanding.

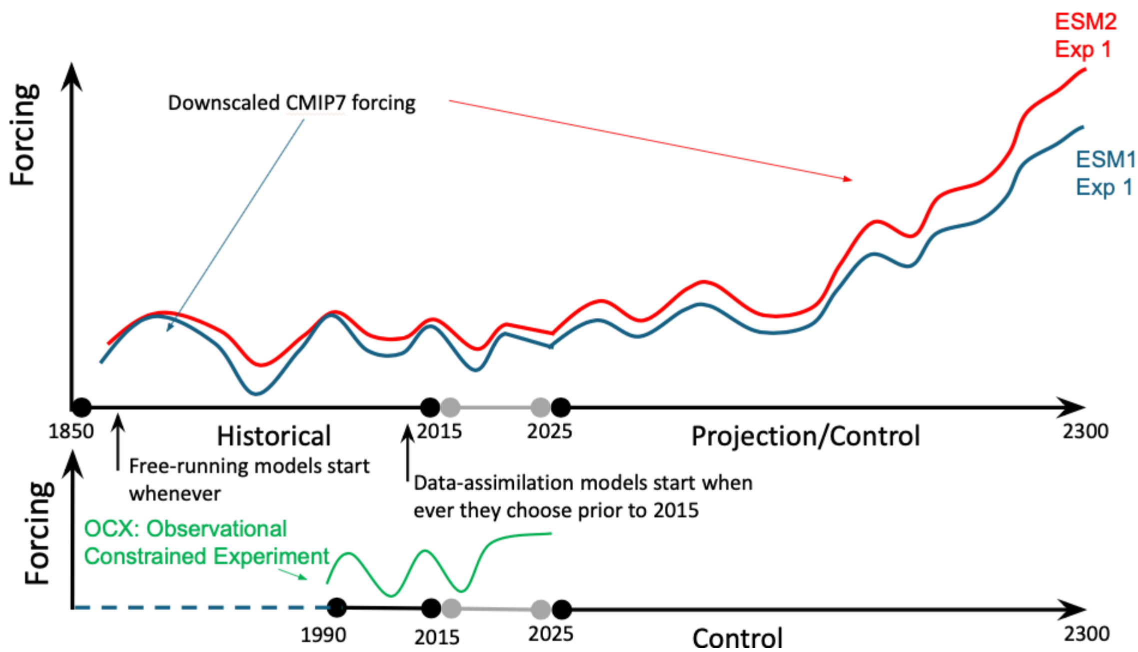


Figure 1: Overview of experimental set up for historical to ISMIP7 projections, and observational constrained experiment.

Greenland Ice Sheet trial protocol

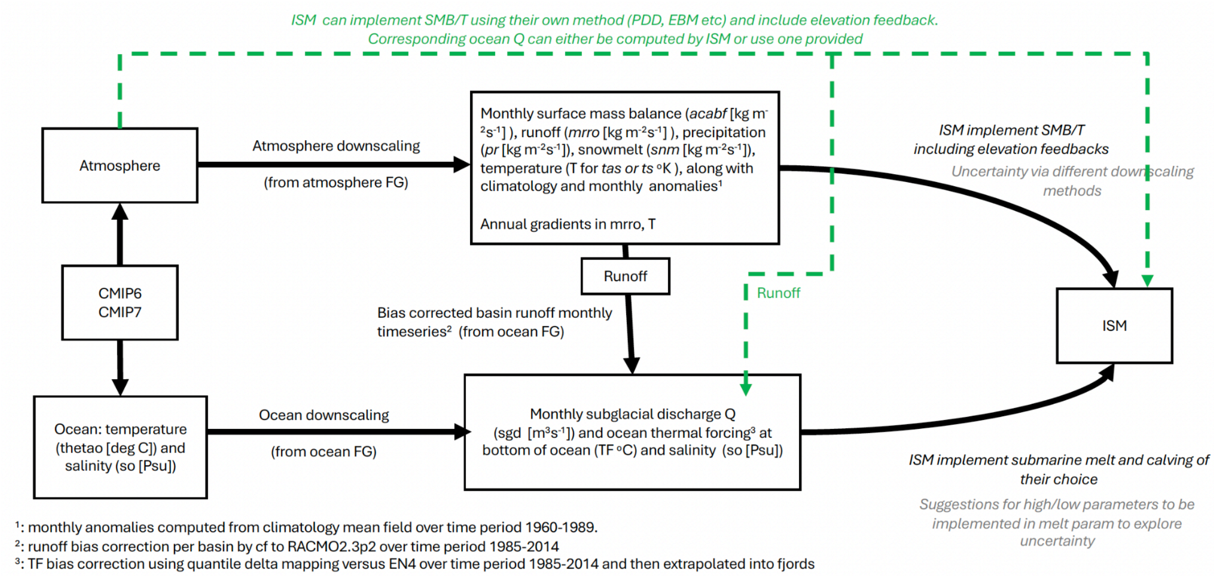


Figure 2: Overview of the ISMIP7 Protocol for the Greenland ice sheet. In short, focus groups provide downscaled atmospheric and oceanic fields, along with a method for sampling the uncertainty and including GIA.

GrIS Atmosphere

Similar to the ISMIP6 protocol, modelers can choose to use the surface mass balance and temperature anomalies, along with their gradient to compute ice-elevation feedback. See ISMIP6 Greenland protocol in Nowicki et al. (2020) or https://theghub.org/groups/ismip6/wiki/ISMIP6-Projections-Greenland

The major difference to ISMIP6 is that now the downscaled SMB and temperature fields are provided as monthly data and as monthly anomalies. The reference period used to calculate the climatology and anomalies is 1960-1989. Depending on the date corresponding to the “initialization”, for example 1900, it may be more appropriate for groups to compute anomalies and a climatology directly from the provided downscaled SMB and temperature data. Please reach out to the FG or SSC if you need advice or additional datasets.

Note that the SMB fields are in units [kg m^{-2} s^{-1}]. To convert to units typically used in an ice sheet model [m yr-1], multiply the variables: SMB [m yr^{-1}] = SMB [kg m^{-2} s^{-1}] * (31556926 [s yr^{-1}] /1000) * (rhof/rhoi) where rhof is freshwater density (1000 [kg m^{-3}]) and rhoi is your specific ice density (typically around 917.0 [kg m^{-3}])

Modelers wishing to use their own PDD or EBM methods can also choose to do so. ISMIP7 will provide the necessary CMIP fields. Depending on your method for GrIS ocean, you may need to compute the runoff for the subglacial discharge and thus modify the GrIS ocean forcing. You can do so using the GrIS ocean scripts. Alternatively, you can also assume that the subglacial discharge would have been the same as the ISMIP7 computed values and use the provided GrIS ocean forcing.

See Atmosphere Focus Group page for links to recording and slides linked with the January 2026 webinar, and the March 2026 Copenhagen workshop.

GrIS Ocean

The protocol for the GrIS ocean was presented in October 2025. See GrIS ocean focus group page for links to recording, slides and documentation. The March 2026 Copenhagen workshop has additional information regarding the exploration of uncertainty.

Groups can implement the submarine melt parameterization of their choice provided that it is driven by the ocean properties from the CMIP models. Modeling group will need to obtain tuning parameters for their melt parameterization (see Copenhagen slides).

Use of a physically based calving process is strongly encouraged.

GrIS Glacial Isostatic Adjustment (GIA)/bedrock adjustment and sea level:

Including GIA/bedrock adjustment in GrIS simulations is optional. There is the opportunity to perform optional simulations evaluating sensitivity of projections to different bedrock adjustment choices. See the GIA focus group page for the Copenhagen presentation and the experimental protocol, which includes an accessible “cheat sheet” for getting started with including bedrock adjustment in ISMIP simulations.

GrIS Output grid

Ice sheet model outputs are to be reported on the standard ISMIP7 grid that is the closest in resolution to a model native grid. The GrIS standard grid is the same as ISMIP6 and is defined on a polar stereographic projection with a standard parallel at 70° N and a central meridian of 45° W (315° E), referenced to the WGS84 datum (EPSG:3413). The grid domain spans from a lower-left cell center at (−720,000 m, −3,450,000 m) to an upper-right cell center at (960,000 m, −570,000 m). At 1 km resolution, the grid consists of 1,681 × 2,881 cells in the x and y directions. Model output may be submitted at resolutions adapted to the native model grid (20, 16, 10, 8, 5, 4, 2, or 1 km) and will be conservatively interpolated to standard grids of 1 km for archiving and 4 km for diagnostic processing.

GrIS Output variables

The ISM data request will largely be based on the ISMIP6 data request (see Appendix of https://theghub.org/groups/ismip6/wiki/ISMIP6-Projections-Greenland). In addition, we would like modelers to save temperature gradients at the base of the ice, and vertically averaged temperature if this is applicable to your model. We may ask a few additional variables. File naming convention may also change slightly to better reflect the uncertainty sampling. We will however continue using one file per variable.

Antarctic Ice Sheet trial protocol

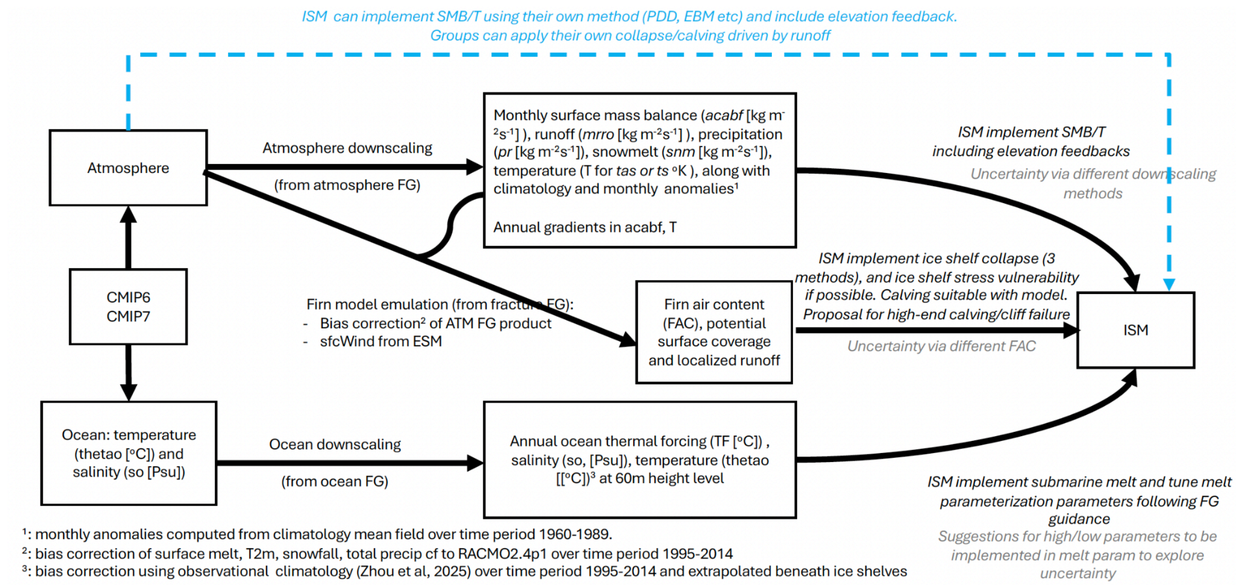


Figure 3: Overview of the ISMIP7 Protocol for the Antarctic ice sheet. In short, focus groups provide downscaled atmospheric, oceanic and fracture fields along with a method for sampling the uncertainty and for including GIA.

AIS Atmosphere

Unlike ISMIP6, the ISMIP7 AIS atmosphere protocol provides downscaled surface mass balance and temperature along with their anomalies and gradient to compute ice-elevation feedback (similar to the protocol for Greenland protocol in Nowicki et al. (2020) or https://theghub.org/groups/ismip6/wiki/ISMIP6-Projections-Greenland)

The major difference to ISMIP6 is that now the downscaled SMB and temperature fields are provided as monthly data and as monthly anomalies. The reference period used to calculate the climatology and anomalies is 1960-1989. Depending on the date corresponding to the “initialization”, for example 1900, it may be more appropriate for groups to compute anomalies and a climatology directly from the provided downscaled SMB and temperature data. Please reach out to the FG or SSC if you need advice or additional datasets.

Note that the SMB fields are in units [kg m^{-2} s^{-1}]. To convert to units typically used in an ice sheet model [m yr-1], multiply the variables: SMB [m yr^{-1}] = SMB [kg m^{-2} s^{-1}] * (31556926 [s yr^{-1}] /1000) * (rhof/rhoi) where rhof is water density (1000 [kg m^{-3}]) and rhoi is your specific ice density (typically 917.0 [kg m^{-3}]).

Modelers wishing to use their own PDD or EBM methods can also choose to do so. ISMIP7 can provide the necessary CMIP fields. Please contact ISMIP7 if this option interests you.

See Atmosphere Focus Group page for links to recording and slides linked with the January 2026 webinar, and the March 2026 Copenhagen workshop.

AIS Ocean

The protocol for the AIS ocean was presented in November 2025 and during the March 2026 Copenhagen workshop. See AIS ocean focus group page for links to recording, slides and documentation.

Groups can implement the submarine ice shelf melt parameterization of their choice provided that it is driven by the ocean properties from the CMIP models. The only recommendation is to avoid linear formulations and to update the ISMIP6 formulation following Burgard et al. (2022). Modeling groups will need to obtain tuning parameters for their melt parameterization (see Copenhagen slides). To facilitate this work, the AIS Focus group has created toolbox.

AIS Fracture and Calving

There are three possible approaches for AIS ice shelf collapse and we also have a protocol for high-end calving. In Path A, ISM use their own scheme for ice shelf collapse driven by the availability of surface runoff. This implementation allows for collapse resulting from the coupling to existing hydrofracture or damage or stress-based criteria that is internal to the ISM. Path B is a "stress aware hydrofacture" in which surface runoff and excess melt is routed into lakes and combined with the ice shelf stress state to determine collapse. The lake depth and area can be estimated via the method of Grau et al. (2025), while the ISM stress conditions follow from any community-adopted stress or damage triggers such as the Nye zero order fracture or the resistive stress pre-conditioning criteria from Lai et al. (2020). In Path C, ice shelf collapse is imposed when an excess melt or runoff threshold has been reached. This third approach and implementation in ISM is the most simple and similar to that of the ISMIP6 protocol. However, unlike the ISMIP6 protocol, it is recommended that ice shelf vulnerability stress conditions of Lai et al. (2020) for example are also included in the ice sheet model implementation, so that ice shelf removal only occurs in regions of excess melt/runoff threshold exceedance that also correspond to regions of stress-based ice shelf vulnerability (Reynolds and Nowicki, 2026). Please see AIS fracture focus group page for links to the webinar and slides presented in January 2026 and during the March 2026 Copenhagen workshop.

AIS Glacial Isostatic Adjustment (GIA)/bedrock adjustment and sea level:
Including GIA/bedrock adjustment in ISM simulations is optional but highly encouraged for Antarctica. There is the opportunity to perform optional simulations evaluating sensitivity of projections to different bedrock adjustment choices. See the GIA focus group page for the Copenhagen presentation and the experimental protocol, which includes an accessible "cheat sheet" for getting started with including bedrock adjustment in ISMIP simulations.

AIS Output grid
Ice sheet model outputs are to be reported on the standard grid that is the closest to a model native grid. The AIS standard grid is the same as ISMIP6 and defined using a polar stereographic projection with a standard parallel at 71° S and a central meridian of 0° W, referenced to the WGS84 datum (EPSG:3031). The grid domain extends from (−3,040,000 m, −3,040,000 m) in the lower-left corner to (3,040,000 m, 3,040,000 m) in the upper-right corner. At the standard 8 km resolution, the grid comprises 761 × 761 cells in the x and y directions. Model output may be submitted at resolutions consistent with the native model grid (32, 16, 8, 4, 2, or 1 km) and will be conservatively interpolated to the standard 8 km grid for archiving and diagnostic processing.

AIS Output variable and grid
The ISM requested data request will largely be based on ISMIP6 data request (see Appendix of https://theghub.org/groups/ismip6/wiki/ISMIP6-Projections2300-Antarctica). In addition, we would like modelers to save temperature gradients at the base of the ice, and mean temperature if this is applicable to your model. We may ask a few additional variables. File naming convention may also change slightly to better reflect the uncertainty sampling. We will however continue using one file per variable.

Deadlines and Support
We have a new discussion board on GitHub https://github.com/orgs/ismip/discussions and will host a series of virtual coffee hours. We also welcome questions by email.

30th June 2026: Preliminary submission of at least one core projection (SSP585 for CESM2-WACCM) and readme with metadata etc and indication of planned ESM/PPE participation

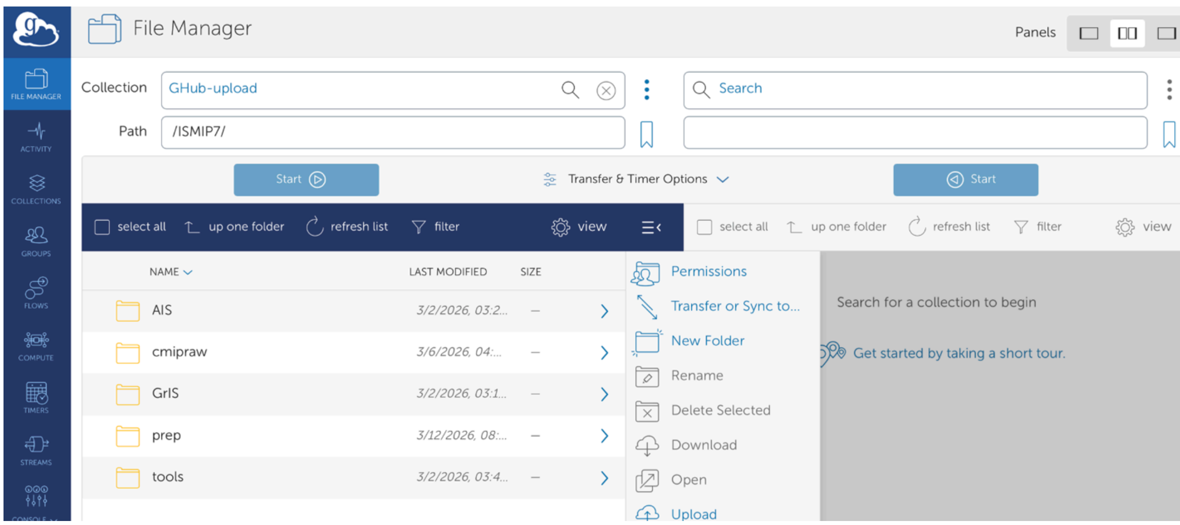
30th September 2026: Full submission of all core experiments and confirmation of planned ESM/PPE participation

31st October 2026: Any submission of ESM/PPE experiments by that date will be included in the group papers

15th March 2027: IPCC submission deadline

Where to obtain ISM forcing dataset?

We use Globus for data distribution. The simplest method is to transfer the data via the globus webpage – see screenshot – but there is also a command line interface (CLI). An example of file upload and download with CLI is below. For help trouble shooting, please visit the Globus documentation.



Notes:

- Some of the Globus links or paths differ slightly from the webinar information, this cheat sheet has the latest information!
- Please use the latest version, denoted with *v2* or *v3* or *v4*. The version for the atmosphere and ocean may differ, as these datasets are prepared independently by the different focus groups.
- For GrIS ocean simulations, the subglacial discharge (Q or sgd) is a product of the atmosphere and will therefore be under the atmosphere folder. This will allow for new Q to be provided with different downscaling methods for the atmosphere.
- You can initiate a transfer via Globus and close your browser/terminal window. Ie: do not worry about terminating a task if it takes longer than you thought - it will continue.

Dataset status - April 2026

- GrIS dataset for CESM2-WACCM historical, ssp126, ssp370, ssp534-over and ssp585 are ready
- AIS dataset for CESM2-WACCM historical and ssp585 are ready

GrIS dataset
(old ISMIP7_prep path here) and new ISMIP7 main directory path:
https://app.globus.org/file-manager?origin_id=ccc9bbd2-4091-4e35-addd-eeb639cf5332&origin_path=%2FISMIP7%2F

Files linked to GrIS Ocean Protocol:
bias-corrected, extrapolated, regridded to horizontal ISMIP grid thermal forcing and salinity
/ISMIP7/GrIS/CESM2-WACCM/[historical/ssp585/..]/**ocean**/[so/tf]/v2/
The subglacial discharge (Q or sgd) is a product of the atmosphere and will therefore be under the atmosphere folder.

Files linked to GrIS Atmosphere Protocol:
Downscaled and regridded to ISMIP grid. Surface mass balance and its component, temperature etc along with climatology and anomalies. Recommendation is to use the anomaly data directly, but this may also depend on the time for your initialization. Climatology is referenced to 1960-1989. If you are using your own PDD scheme or EMB scheme, please contact us so that we can provide you the files that you need.
/ISMIP7/GrIS/CESM2-WACCM/[historical/ssp585/..]/**SDBN1**/[acabf/dacabfdz/acabf_anomaly/..]/v2/
/ISMIP7/GrIS/CESM2-WACCM/[historical/ssp585/..]/**SDBN1**/[sgd]/v2/

AIS dataset
(old ISMIP7_prep path here) and new ISMIP7 path:
https://app.globus.org/file-manager?origin_id=ccc9bbd2-4091-4e35-addd-eeb639cf5332&origin_path=%2FISMIP7%2F

Files linked to AIS Atmosphere Protocol:
Downscaled and regridded to ISMIP grid. Surface mass balance and its component, temperature etc along with climatology and anomalies. Recommendation is to use the anomaly data directly, but this may also depend on the time for your initialization. Climatology is referenced to 1960-1989. If you are using your own PDD scheme or EMB scheme, please contact us so that we can provide you the files that you need
/ISMIP7/AIS/CESM2-WACCM/[historical/ssp585/..]/**SDBN1**/[acabf/dacabfdz/acabf_anomaly/..]/v2/

Files linked to AIS Ocean Protocol:
bias-corrected, extrapolated, regridded to 60m vertical 8km horizontal ISMIP grid thermal driving, potential temperature, practical salinity
e.g.,/ISMIP7/AIS/CESM2-WACCM/[historical/ssp585/..]/ocean/[so/tf/thetao]/v3/

Recommendation is to use the data directly, as it is bias-corrected to the latest climatology of the Southern Ocean (Zhou et al., in review), but anomalies are OK. Climatology is referenced to 1995-2024
e.g.,/ISMIP7/AIS/obs/ocean/climatology/zhou_annual_06_nov/[so/tf/thetao]/v4/

Additional files linked to AIS Ocean Protocol:
Other small datasets on Globus that maybe needed for the AIS ocean
Basins:
/ISMIP7/AIS/obs/ocean/IMBIE-basins/v3/IMBIE-basins_AIS_obs_ocean_v3.nc
Ocean grid:
/ISMIP7/AIS/grid/ocean/ISMIP7/8km-60m/v3/ISMIP7_8km-60_AIS_grid_ocean_v3.nc
Topography:
/ISMIP7/AIS/obs/ocean/topography/
bedmap3_AIS_obs_ocean_topography_v3.nc
BedMachineAntarctica-v3_AIS_obs_ocean_topography_v3.nc
Buttressing regions:
/ISMIP7/AIS/parameterisations/ocean/bfrn/
BFRN_ismip_8km.nc, other resos
Floating masks:
/ISMIP7/AIS/parameterisations/ocean/floatingmasks/
BFRN_ismip_8km.nc, other resos
Calibration data from ocean models:
/ISMIP7/AIS/parameterisations/ocean/ocean_modeling_data
melt_cold_target_term3.nc
melt_warm_target_term3.nc
Mathiot23_cold_T/S/TF/m.nc
Mathiod23_warm_T/S/TF/m.nc
TimmermannUndGoeller2017_cold_T/S/TF/m.nc
TimmermannUndGoeller2017_warm_T/S/TF/m.nc
[....]
Calibration data from ocean observations:
/ISMIP7/AIS/parameterisations/ocean/ocean_observation_data
melt_observations_target_term4.nc
Dutrieux_ismip8km_60m_*.nc
[...]
Calibration data from melt observations:
/ISMIP7/AIS/parameterisations/ocean/meltobs
melt_target_term2.nc
melt_paolo_err_adusumilli_ismip8km.nc
Melt_Paolo_Err_Adusumilli_imbie2_v2.csv

MeltMIP datasets:
/ISMIP7/AIS/meltMIP

Files linked to AIS Fracture Protocol:
Excess melt datasets:
/ISMIP7/AIS/CESM2-WACCM/[historical/ssp585/..]/**fracture**/
OTHER FILES COMING SOON

Files linked to Global CMIPraw dataset
cmipraw dataset are the original CMIP datasets downloaded from ESGF. These files maybe useful for modelers with their own PDD or EBM methods, but please contact us if you decide to use these.

Files linked to Greenland and Antarctic Observational MIPkit:
https://theghub.org/dataset-listing or in the respective ice sheet directories.
/ISMIP7/[AIS/GrIS]/obs

GLOBUS CLI Examples

Example of command line File transfer to GLOBUS GHub-upload:

```
# GLOBUS commandline transfer (from a linux server):
cd /scratchu/mydirectory/ISMIP7_GREEN_EXPERIMENT
chmod a+rx *.nc
pip install globus-cli
globus login # follow instructions
globus whoami
globus endpoint search "ghub" # shows ID of enpoint called "GHub-upload"
globus ls ccc9bbd2-4091-4e35-addd-eeb639cf5332
globus ls ccc9bbd2-4091-4e35-addd-eeb639cf5332/ISMIP6
globus ls ccc9bbd2-4091-4e35-addd-eeb639cf5332:ISMIP6/ISMIP7_Prep/AIS_ocean/contemporary
# then, need to define the local endpoint:
wget https://downloads.globus.org/globus-connect-personal/linux/stable/globusconnectpersonal-latest.tgz
cd globusconnectpersonal-3.2.8
./globusconnectpersonal -setup --no-gui
./globusconnectpersonal -start &
globus endpoint local-id  # shows ID of local endpoint
vi ~/.globusonline/lta/config-paths # add line to allow path: /scratchu/mydirectory/ISMIP7_GREEN_EXPERIMENT/,1,0
globus transfer
005e08c1-f000-00f0-afb1-0e56852e4f6b:/scratchu/mydirectory/ISMIP7_GREEN_EXPERIMENT/tf_Oyr_contemporary_most_ismip8km_60m_1950-2025.nc
ccc9bbd2-4091-4e35-addd-eeb639cf5332:ISMIP6/ISMIP7_Prep/AIS_ocean/contemporary/tf_Oyr_contemporary_most_ismip8km_60m_1950-2025.nc
```

Example of command line File transfer from GLOBUS GHub-upload:

```
# GLOBUS commandline transfer (from GHub-upload to your linux server):
pip install globus-cli
globus login
globus whoami
globus endpoint search "ghub"
```

```
globus ls ccc9bbd2-4091-4e35-addd-eeb639cf5332
globus ls ccc9bbd2-4091-4e35-addd-eeb639cf5332/ISMIP6
globus ls ccc9bbd2-4091-4e35-addd-eeb639cf5332:ISMIP6/ISMIP7_Prep/AIS_ocean/contemporary

wget https://downloads.globus.org/globus-connect-personal/linux/stable/globusconnectpersonal-latest.tgz
tar xvf globusconnectpersonal-latest.tgz
cd globusconnectpersonal-3.2.8
./globusconnectpersonal -setup --no-gui
./globusconnectpersonal -start &
globus endpoint local-id
vi ~/.globusonline/lta/config-paths
# Add: /scratchu/mydirectory/Downloads_from_GHub/,1,0

globus transfer \
ccc9bbd2-4091-4e35-addd-eeb639cf5332:ISMIP6/ISMIP7_Prep/AIS_ocean/contemporary/tf_Oyr_contemporary_most_ismip8km_60m_1950-2025.nc \
005e08c1-f000-00f0-afb1-0e56852e4f6b:/scratchu/mydirectory/Downloads_from_GHub/tf_Oyr_contemporary_most_ismip8km_60m_1950-2025.nc
globus task list
```

GLOBUS upload/download comparison table:

Operation	Source Endpoint → Destination Endpoint	Example Command	Notes
Upload	Local GCP → GHub_upload	globus transfer <LOCAL_ID>:/path/file <REMOTE_ID>:/path/file	Local directory must be readable.
Download	GHub_upload → Local GCP	globus transfer <REMOTE_ID>:/path/file <LOCAL_ID>:/path/file	Local directory must be writable.
Upload (Recursive)	Local directory → Remote directory	globus transfer --recursive <LOCAL_ID>:/dir <REMOTE_ID>:/dir	Preserves directory structure.
Download (Recursive)	Remote directory → Local directory	globus transfer --recursive <REMOTE_ID>:/dir <LOCAL_ID>:/dir	Requires write permissions in config-paths.
Browse Remote	N/A	globus ls <REMOTE_ID>:/path/	Useful before transfers.
Browse Local	N/A	globus ls <LOCAL_ID>:/path/	Requires GCP running.
Monitor Tasks	N/A	globus task list / globus task show <TASK_ID>	Applies to both upload and download.